

Yichen Cai

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► **Objective:** Experienced Master student in CS trying to find a research internship that allows me to utilize my skillset in machine learning, computer systems and software engineering to deliver state-of-the-art software products and solutions.

EDUCATION

University of Toronto (St. George), MSc in Applied Computing, Canada **Sept 2024 – Dec 2025 (Expected)**
Master of Science in Applied Computing. Computer Science Stream.

University of Toronto (St. George), Computer Science Specialist, Canada **Sept 2019 - June 2024**
Honours Bachelor of Science. Specialized in Artificial intelligence (AI). **GPA 3.99/4.0**
Awards: Dean's List Scholar 2020, 2021, 2022 and 2024. University of Toronto Scholar.

PROFESSIONAL EXPERIENCE

Software Engineer - Zebra Technologies (Mississauga, ON.) **May 2022 - July 2023**

- Zebra System Applications - Main Developer:
 - Created Zebra Pass-Through Antenna setting app, enabling 2 modes of communication with Zebra dock station.
 - Developed the heater control app that regulates device temperatures, making devices suitable for frozen warehouses.
 - Collaborated with the PM and hardware team (AGILE) to use the GPIO drivers to control the antenna and heaters.
 - Efficiently stored the user configurations to minimize processing time.

Software Engineer - Orali Mobile (Toronto, ON.) **Sept 2021 - Dec 2021**

- Orali Money App - Developer:
 - Designed and implemented an Android money wallet app using the MVVM architecture with Java.
 - Aimed to enable the 1 billion financially illiterate African population to engage in formal economic activities.
 - Established methods to communicate with the local banking system through USSD using the Hover APIs.
 - Created an automated testing/deployment system through CI/CD tools to continuously test and update the app.

PROJECTS & PUBLICATIONS

Key-Value Database Design and Implementation (Toronto, ON.) **July 2023 - Oct 2023**

- Designed a full KV database with mem-table, SST storage and extendible-hashing buffer pool in C++.
- Created an AVL-tree based mem-table for fastest fetching of data.
- Optimized the performance of the database by indexing the persistent SSTs with B-tree structure.
- Supported efficient point queries and range queries using buffer pool w/ LRU policy.
- The overall performance is comparable to mainstream KV databases like RockDB.

Team Lead - NVIDIA AICity Challenge Project, CVPR 2023 (Toronto, ON.) **Dec 2022 - June 2023**

- Retail Product Recognition and Counting System:
 - Created a project plan and scheduled tasks to ensure project completion within deadlines.
 - Designed, implemented and trained a YOLO + StrongSORT backbone algorithm.
 - Fine-tuned the model to recognize 116 distinct products from retail stores with an F1-score of 0.82.
 - Achieved top 3 finish at CVPR 2023 NVIDIA AICity Challenge, published in CVPR 2023 proceedings ([DACNet](#)).

CS SKILL SETS

- **Programming languages:** Python (w/ Pytorch, TF), C/C++, Java, SQL, Shell.
- **Tools:** Git, CI/CD, GCP, JetBrains.